

Math 12 Curriculum

Intermediate Algebra | Middlesex School Mathematics | OpenStax: OpenStax Viewer

COURSE AT A GLANCE

Review Content

REVIEW

Primary: [1.1 Use the Language of Algebra](#); [1.3 Fractions](#); [1.5 Properties of Real Numbers](#)
Supplemental: [1.2 Integers](#)

UNIT LEARNING OBJECTIVES

- I can evaluate numerical expressions using the order of operations.
- I can perform operations with signed integers and fractions.
- I can simplify an algebraic expression by combining like terms.
- I can use the distributive property to rewrite an expression.
- I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

Primary: [2.1 Use a General Strategy to Solve Linear Equations](#); [2.3 Solve a Formula for a Specific Variable](#); [2.2 Use a Problem Solving Strategy](#); [2.4 Solve Mixture and Uniform Motion Applications](#)
Supplemental: [2.2 Use a Problem Solving Strategy](#)

UNIT LEARNING OBJECTIVES

- I can solve a linear equation and present equivalent steps clearly.
- I can solve a linear equation containing fractions.
- I can rearrange a formula to isolate a specified variable.
- I can translate a verbal relationship into a linear equation.
- I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

Primary: [2.5 Solve Linear Inequalities](#); [2.6 Solve Compound Inequalities](#); [2.7 Solve Absolute Value Inequalities](#)

UNIT LEARNING OBJECTIVES

- I can solve and graph a one-variable linear inequality.
- I can solve a compound inequality involving AND.
- I can solve a compound inequality involving OR.
- I can express an inequality solution using interval notation.
- I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

Primary: [3.2 Slope of a Line](#); [3.1 Graph Linear Equations in Two Variables](#); [3.3 Find the Equation of a Line](#); [3.5 Relations and Functions](#)

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation in a common form.
- I can write an equation of a line from given information.
- I can determine whether two lines are parallel, perpendicular, or neither.
- I can evaluate a function using function notation.
- I can determine whether a relation is a function using the vertical line test or input-output data.
- I can determine domain and range from a graph or table.
- I can interpret domain and range in context.

Systems of Linear Equations

REQUIRED

Primary: [4.1 Solve Systems of Linear Equations with Two Variables](#); [4.2 Solve Applications with Systems of Equations](#)

UNIT LEARNING OBJECTIVES

I can interpret the solution of a linear system as an intersection point.

I can solve a two-variable linear system by graphing.

I can solve a two-variable linear system by substitution.

I can solve a two-variable linear system by elimination.

I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

Primary: [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [5.4 Dividing Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#)

UNIT LEARNING OBJECTIVES

I can simplify expressions using integer exponent rules.

I can add and subtract polynomials.

I can multiply polynomials, including binomials.

I can divide a polynomial by a monomial.

I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

Primary: [Intermediate Algebra 2e 2.7 Solve Absolute Value Inequalities](#); [Intermediate Algebra 2e 4.7 Graphing Systems of Linear Inequalities](#); [Intermediate Algebra 2e 6.2 Factor Trinomials](#); [Precalculus 2e 1.2 Domain and Range](#)

UNIT LEARNING OBJECTIVES

I can solve and graph an absolute-value inequality.

I can graph a system of linear inequalities and identify its solution region.

I can factor a quadratic trinomial.

I can identify a relationship as discrete or continuous.